



4.2.3 Solving the normal equations ¶

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fit width



Let us review a method you have likely seen before for solving the LLS problem when matrix A has linearly independent columns. We already used these results in [Subsection 2.1.1](#)

We wish to solve $A^H A \hat{x} = A^H b$, where A has linearly independent columns. If we form $B = A^H A$ and $y = A^H b$, we can instead solve $B \hat{x} = y$. Some observations:

- Since A has linearly independent columns, B is nonsingular. Hence, $\hat{x}$ is unique.
- B is Hermitian since $B^H = (A^H A)^H = A^H (A^H)^H = A^H A = B$.
- B is Hermitian Positive Definite (HPD): $x \neq 0$ implies that $x^H B x > 0$. This follows from the fact that

$$x^H B x = x^H A^H A x = (Ax)^H (Ax) = \|Ax\|_2^2.$$

Since A has linearly independent columns, $x \neq 0$ implies that $Ax \neq 0$ and hence $\|Ax\|_2^2 > 0$.

In [Section 5.4](#), you will find out that since B is HPD, there exists a lower triangular matrix L such that $B = LL^H$. This is known as the Cholesky factorization of B . The steps for solving the normal equations then become

- Compute $B = A^H A$.

Notice that since B is Hermitian symmetric, only the lower or upper triangular part needs to be computed. This is known as a Hermitian rank- k update (where in this case $k = n$). The cost is, approximately, mn^2 flops. (See [Section C.1](#).)

- Compute $y = A^H b$.

The cost of this matrix-vector multiplication is, approximately, $2mn$ flops. (See [Section C.1](#).)

- Compute the Cholesky factorization $B \rightarrow LL^H$.

Later we will see that this costs, approximately, $\frac{1}{3}n^3$ flops. (See [Subsection 5.4.3](#).)

- Solve

$$Lz = y$$

(solve with a lower triangular matrix) followed by

$$L^H \hat{x} = z$$

(solve with an upper triangular matrix).

Together, these triangular solves cost, approximately, $2n^2$ flops. (See [Section C.1](#).)

We will revisit this in [Section 5.4](#).